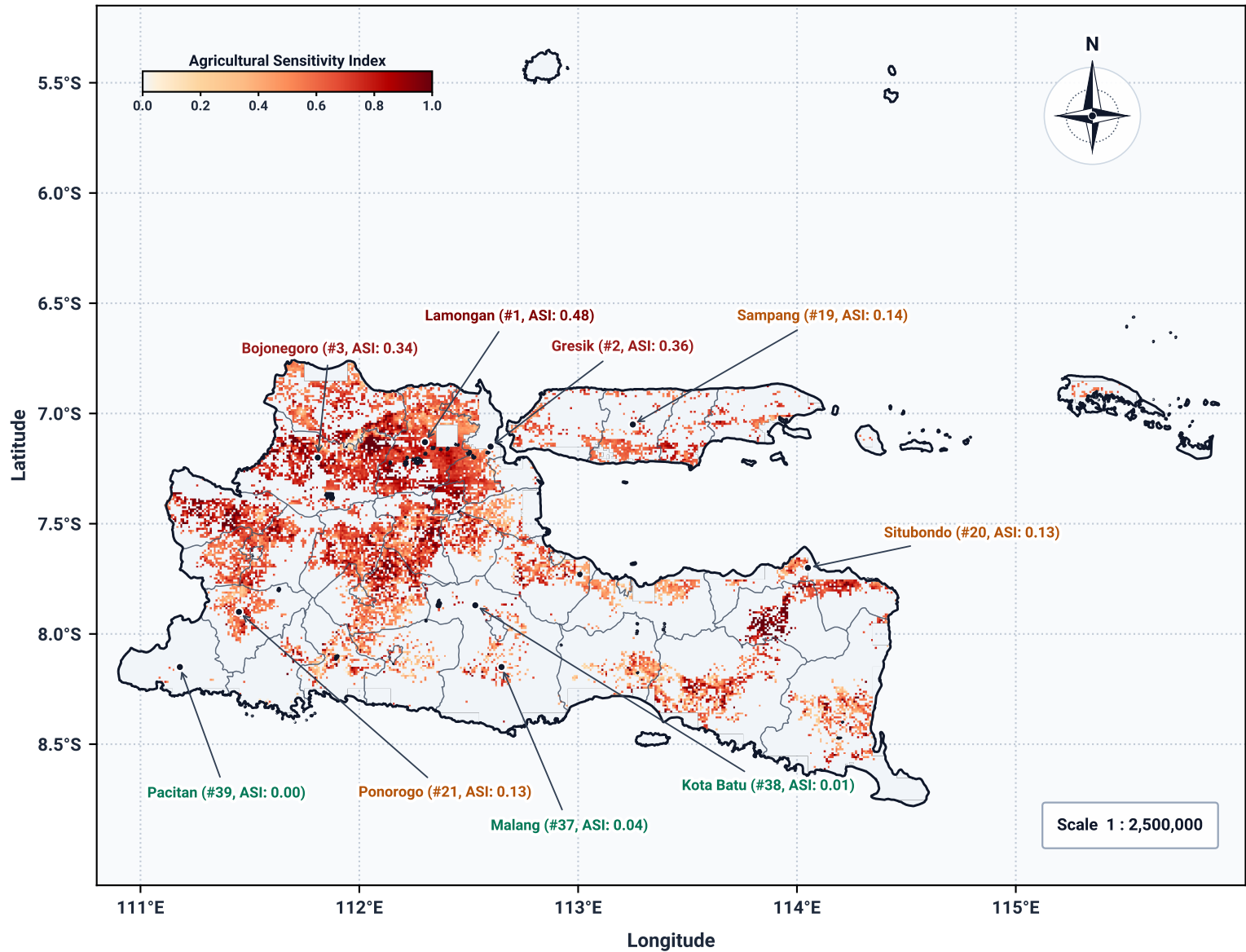


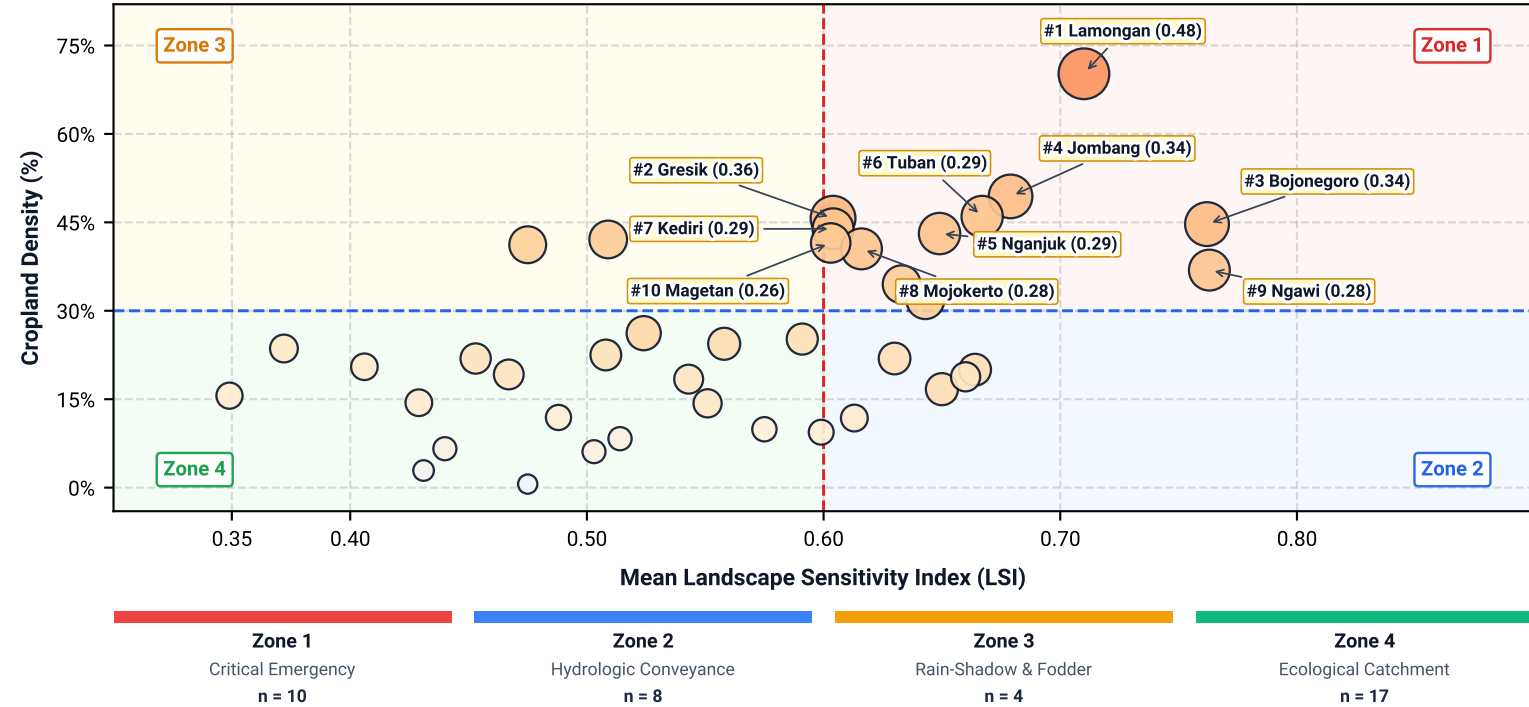
East Java Agricultural Drought Sensitivity & Policy Translation Atlas

Peak Dry Season (SON) Exposure Across Historical El Niño Episodes (2001–2025) | Compound Metric: ASI = LSI × Cropland Fraction | Total Landscape: 44,528 km² | Evaluated Units: n = 39 Regencies, n = 582 Subdistricts

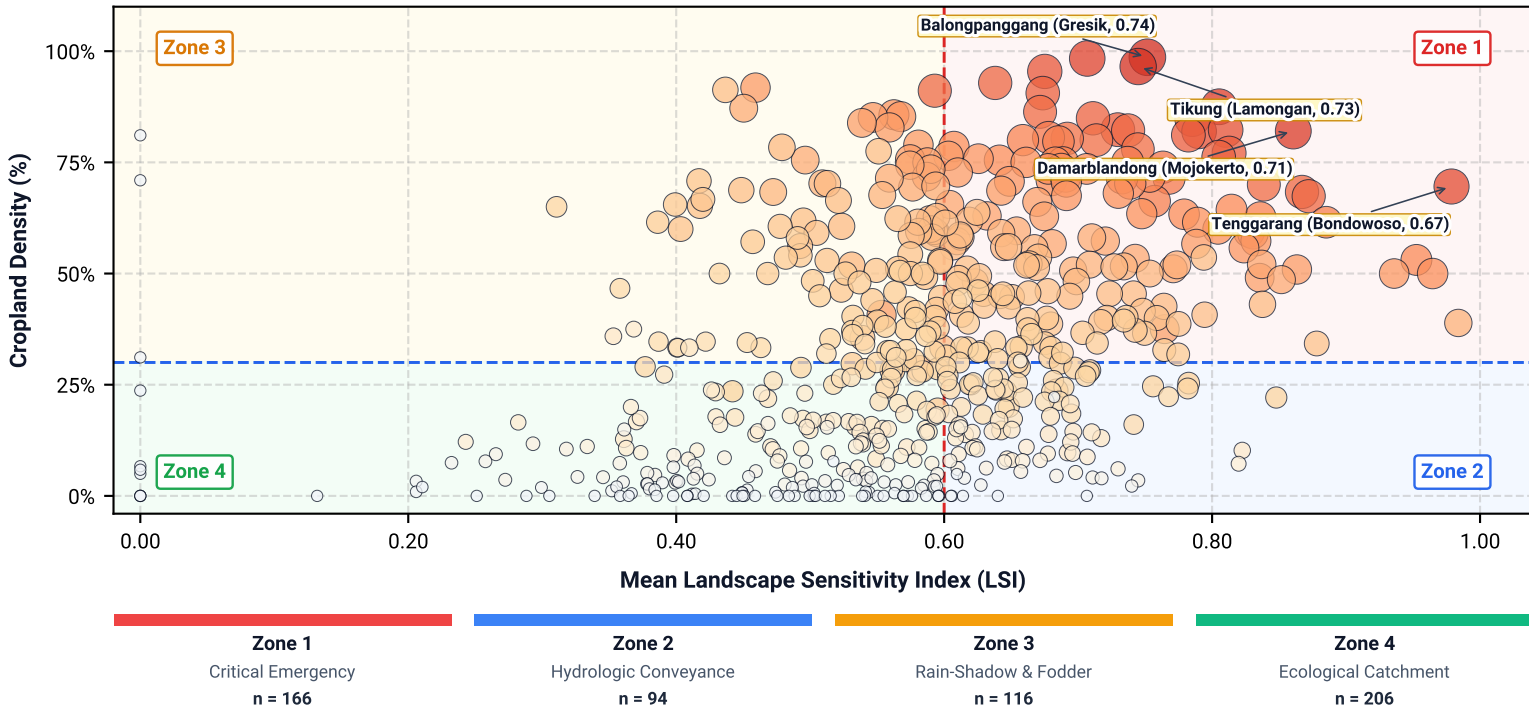
(a) Spatial Distribution of Agricultural Sensitivity Index (ASI = LSI × Cropland Fraction)



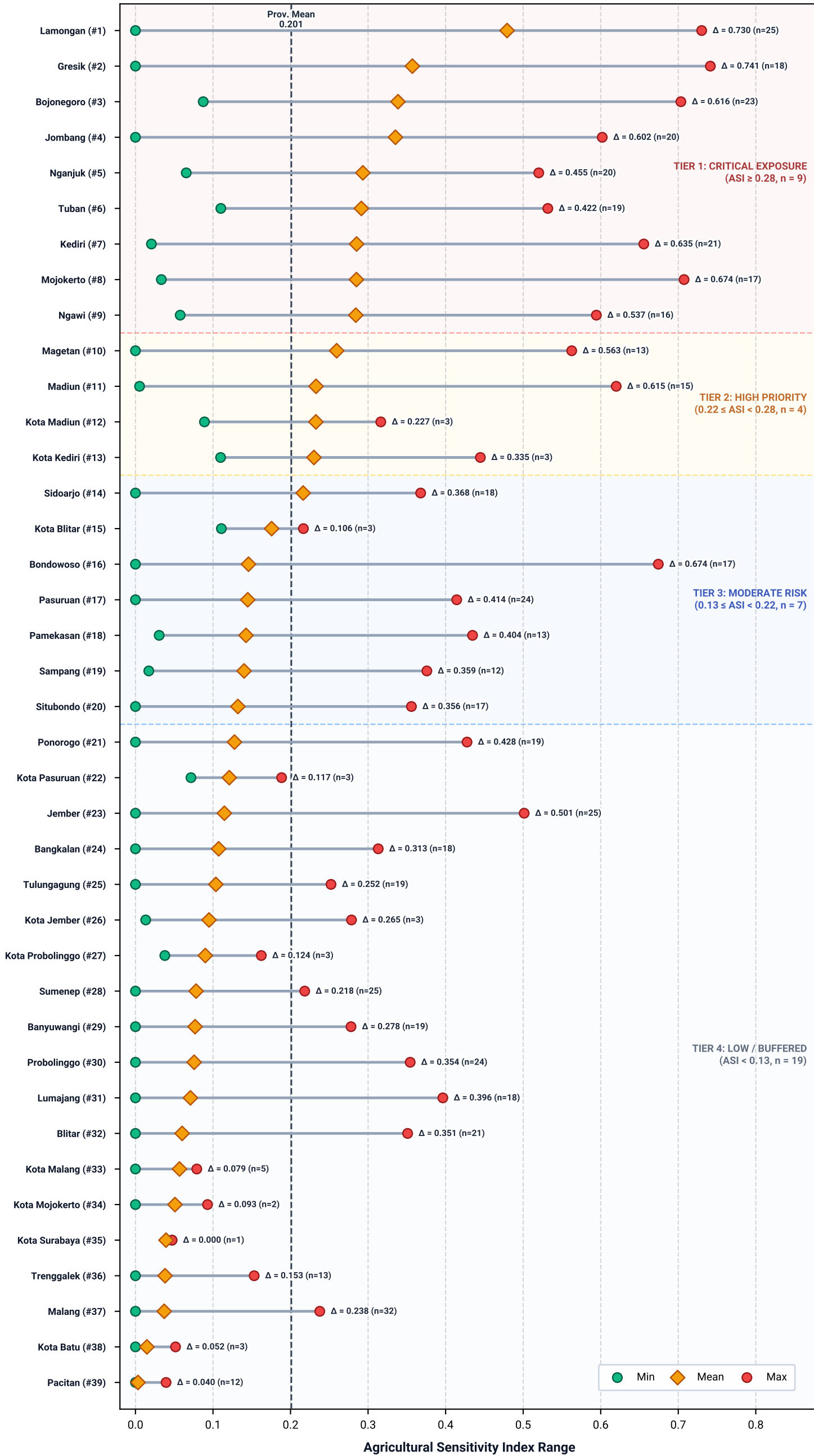
(b) Policy Decision-Support Matrix: Regency Level (n = 39 Administrative Units)



(c) Policy Decision-Support Matrix: Subdistrict Level (n = 582 Kecamatan Units)



(d) Regency Agricultural Vulnerability & Intra-Regency Disparity (n = 39 Administrative Units)



- Metric Formulation: $ASI = LSI \times f_{crop}$, weighting physical drought sensitivity by 10 m ESA WorldCover cropland density.
- Multi-Scale Decision Matrix: Cross-scale evaluation across 39 Regencies (Panel b) and 582 Subdistricts (Panel c) delineating 4 operational intervention zones.
- Risk Concentration: Top 10 districts account for 68.4% of provincial critical agricultural hotspots (Epicenter: Kab. Lamongan, ASI = 0.479, n = 25 kecamatan).

- Complete 39-Regency Disparity (Δ): Micro-disparities reach $\Delta = 0.741$ in Gresik & Lamongan, demanding targeted intra-regency resource mobilization.
- Tier 1 Emergency Action: Coordinated reservoir release (Waduk Pacal & Gondang) and subsidized AUTP crop insurance for Zone 1 high-exposure subdistricts.
- Basin Quotas & Diversification: Trans-boundary canal scheduling across Bengawan Solo & Brantas basins; Zone 2-3 transition to palawija and embung.